

The Human Factor in Cybersecurity

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Introduction

Human factors have a very big impact on cybersecurity, due to the fact that people interact with systems, data, emails, and security control every single day. Even if organizations have strong cyber technical defenses, human decisions can still give a gap for attackers to squeeze in an attack. With that being said cybersecurity should address not only technology, but also user behavior, training, policies, and organizational culture.

Human Factor in Cybersecurity

For me, the human factor is how we affect cybersecurity whether it be the decisions we make, our habits, mistakes, and interactions with technology. Organizations and their products or IT team are responsible for the creation of accounts, users, handling sensitive information, and following security procedures. You would think that it always requires a very technical vulnerability to infiltrate these systems, sometimes an attacker can social engineer an employee and take advantage of whatever comes out of it and use that information to start infiltration.

Human Behavior and its Vulnerability

Some examples we mentioned in past modules are phishing and social engineering. An attacker may send emails that appear to be trusted and convince the said employee to click a link, open the attachment and or provide their credentials or whatever important information was asked. This attack shows how it is heavily relied on human behaviors, like decision-making. This also shows that sometimes urgency on emails creates somewhat an illusion of legitimacy. So with that you can understand that technology systems may work like how they were designed but then the attacker is more likely to find entry points using human mistakes or predicting human behavior.

The second biggest risk in my opinion is creating and reusing weak passwords. With how technology has developed these past years and with AI, weak passwords can be predictable. Failing to protect credentials can make it easier for attackers to gain unauthorized access. Weak passwords can become even more serious when one compromised password is reused across multiple accounts. Organizations can avoid these things through cyber education, MFAs, password requirements and access control lists. They of course do not eliminate human error but they can at least reduce the damage or prevent it at least.

Organizational and Psychological Factors

Cybersecurity behavior is also influenced by psychological and organizational factors. Employees may prioritize getting their work done quickly, especially when security procedures seem inconvenient or slow down normal tasks. Social pressure can also influence decisions. Because of this, organizations should build a culture where employees understand that they are part of the cybersecurity process. Regular, practical training should be consistent with what they can possibly encounter. Policies should be clear and realistic. Ultimately, the goal should be to encourage employees to report mistakes and suspicious activity rather than hide the fear or punishment.

Reducing the Human Related Cybersecurity Risks

I strongly believe that organizations can reduce the threats and risks through cybersecurity education. Regular security awareness training should touch on phishing, social engineering, password security, and proper handling of sensitive information. Organizations need to implement MFA, least privilege, access controls, and continuous monitoring especially when users make a mistake. This is to understand that humans are not perfect and we make mistakes and our behaviors can be predictable which attackers tend to exploit.

Conclusion

At the end of the day human factors play a major role in cybersecurity, every day decisions are made by humans, and with that it either increases or decreases the security risks. That's why phishing, smishing, social engineering, and other forms of "human hacking" have been invented to take advantage of human behaviors to gain access and infiltrate sensitive information and data systems. Organizations can address these risks of course through consistent cyber training, creating clear policies, strong authentication and authorization, and developing a culture that is cyber aware of these potential threats. Throughout the modules and being in the field of cybersecurity, a strong cybersecurity program I believe requires both effective technology and better informed people.